

Group Project 2 Report – Exploring Other Cloud Providers

Course: CS 4843 Cloud Computing

Group Number: 3

Active Members: Declan Petershack Pbb579, Carlos Martinez fxv013, Elric Dominguez tbu965, Fouad Farhoud idu077

Selected Cloud Providers: Microsoft Azure, Google Cloud Platform, IBM Cloud, Oracle Cloud

1. Introduction

Briefly explain which cloud providers you selected and how/why you assigned them to members. Was there any conflict to resolve? If so, how did you resolve it?

The providers we selected are Microsoft Azure, Google Cloud Platform, IBM Cloud, Oracle Cloud. We really just self assigned which ones each person was interested in. There wasn't really any conflict at all. The only thing that came up is we had to add our fourth member to the group but Elric reached out and we got him added.

2. Individual Exploration (experiences, findings, screenshots)

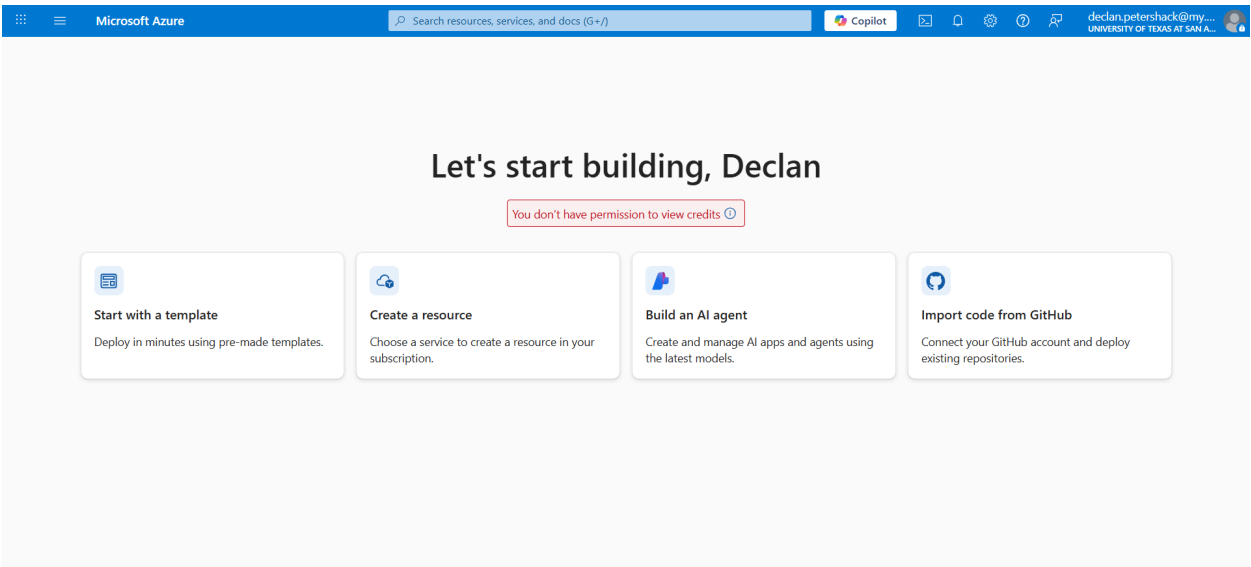
Microsoft Azure

Student 1 Declan Petershack Pbb579 – Microsoft Azure

Briefly describe following based on your experience with the assigned cloud provider:

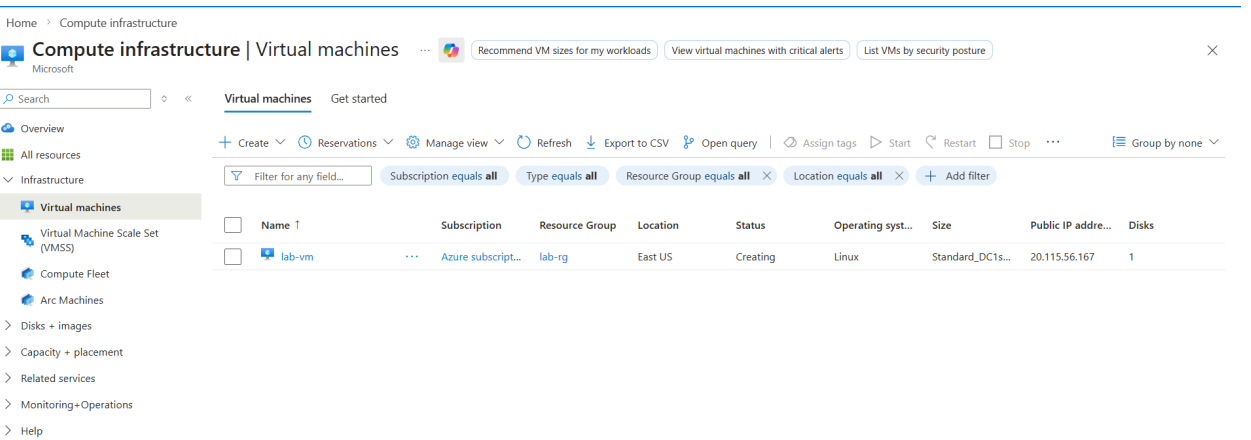
•Account setup experience:

Signing up for Azure needed a payment method, it was pretty fast and I could use my school email for the account. I was able to access the Azure dashboard immediately after logging in.



•VM deployment steps (OS, web server, webpage):

I created a VM using Ubuntu 24.04 LTS on the smallest available size. I connected via SSH, installed Apache, and created a simple webpage.



Hello I am Declan Petershack

My UTSA ID is pbb579
I am part of group 3

• *Basic security setup (firewall/network rules) and its terminology:*

I used Azure network security rules to allow HTTP (80) and SSH (22), making the web server accessible while keeping the VM secure.

The screenshot shows the configuration page for a Network Security Group (NSG) named 'lab-vm-nsg'. It is attached to the network interface 'lab-vm259_z1' and impacts 0 subnets and 1 network interface. The page includes a search bar and filters for Source, Destination, Protocol, Action, and Port. The 'Inbound port rules' section is expanded, showing 5 rules:

Prio...	Name	Port	Protocol	Source	Destination
300	SSH	22	TCP	Any	Any
320	HTTP	80	TCP	Any	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowAzureLoadBalancerInB...	Any	Any	AzureLoadBalancer	Any
65500	DenyAllInBound	Any	Any	Any	Any

The 'Outbound port rules' section is collapsed, showing 3 rules.

The screenshot shows the Microsoft Azure Resource Manager console. The main area displays 'No resource groups to display' with a 'Create' button and a 'Learn more' link. The left sidebar shows the 'Resource Manager' section with options for 'All resources', 'Favorite resources', 'Recent resources', and 'Resource groups'. The right sidebar shows a 'Notifications' panel with a message: 'Deleted resource group lab-rg'.

• Free tier notes and cost awareness (e.g., specific limits, what costs money beyond free tier, what monitoring/billing alert services the provider offers to avoid unexpected charges):

B-series free-tier VMs weren't available for my UTSA account, so I had to use the smallest available VM and deleted it immediately after to avoid charges. Azure also has budget alerts to monitor my spending. I tried using my personal email to see if I could get the free tiers but it would not allow me to use my personal email.

• Challenges encountered and resolution:

I tried using my personal email but it wouldn't let me so I switched to my school email, B-series VMs didn't appear, and I initially got errors about Trusted Launch. I switched Security type to Standard and picked an available VM size.

• Include Screenshots (minimum 5):

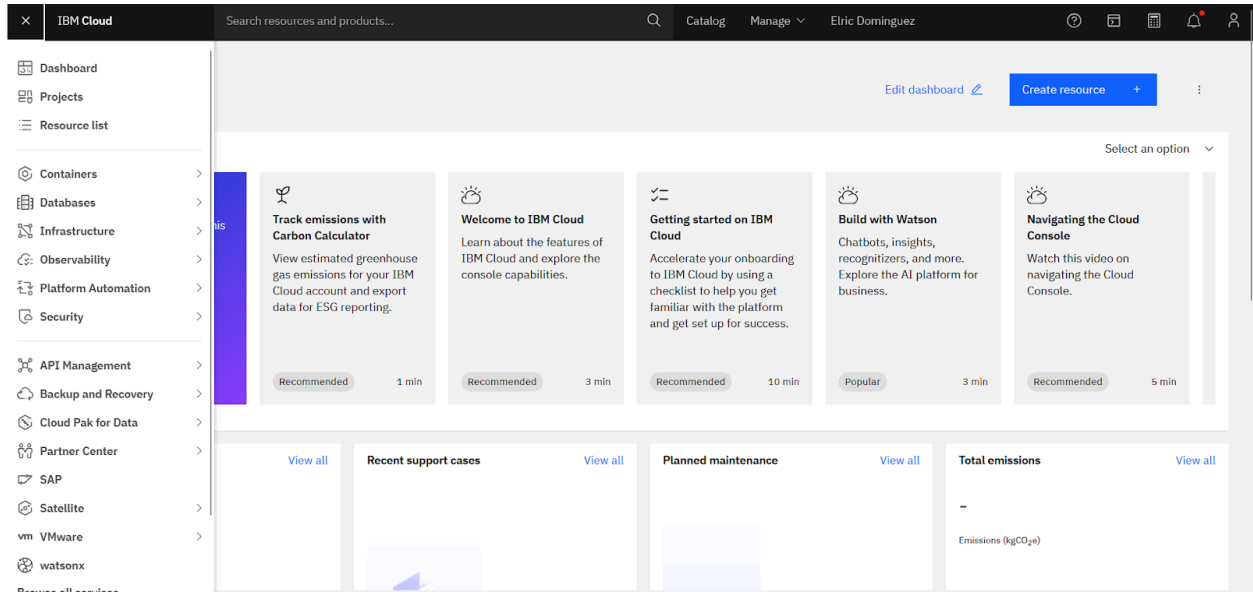
- 1. Cloud console after login
- 2-3. VM instance details page and Webpage running in browser
- 4. Security/firewall rules
- 5. Confirmation of resource termination

Student 2 Elric Dominguez and tbu965 – IBM Cloud

1. Free Account Setup

Signing up for IBM Cloud required a credit card for verification, but the process was fast and straightforward, especially since I used Google to authenticate. The account activated in under 30 seconds with no confusing steps, and I was able to access the IBM Cloud dashboard immediately after logging in.

The only issue is that despite changing my display name to my abc123, my dashboard still displayed my real name.



2. VM Deployment and a static web page

- Note:

The biggest challenges came from the network configuration inside IBM Cloud's security groups. At first, my SSH connection kept timing out because the default security group only allowed traffic from within the group itself. I had to manually create an inbound rule for TCP port 22 to open the VM to external SSH access. Later, when Apache was running but the webpage wouldn't load in the browser, I realized HTTP traffic was also being blocked. Adding a second inbound rule for TCP port 80 allowed public web traffic to reach the server. Once both ports were properly configured, everything worked as expected, and the webpage became accessible from the internet.

VM instance details page (showing OS, size, status, public IP address):

Instance details

tbu965

Virtual server instance ID	Resource group	Location
0727_80e8df03-5ded-4ddd-a6-75-6c8a199c3729	Default	us-south-2
Created date	Virtual private cloud	Provisioned SSH keys
March 25, 2026 at 5:46:05 PM	tbu965-vpc	tbu965

Image details

Ubuntu Linux 22.04 LTS Jammy Jellyfish Minimal Install (amd64)

Image	Image type	Version
ibm-ubuntu-22-04-5-minimal-amd64-13	Stock	22.04 LTS Jammy Jellyfish Minimal Install
Image ID	Architecture	Image status
r006-d15adf1f-27d3-4cf6-8f4e-5fa535bc4616	x86_64	Available

Monitoring preview

Preview based on average of the totals for each metric over the previous 48 hours. For more information, see [Monitoring](#).

CPU usage

3.14%

Memory used

50.2%

[Add monitoring](#)

What do you want to do next?

After you have deployed a virtual server instance, you can manage the server, integrate additional services, and request quota increases.

[Learn about managing your virtual server](#)

[Request quota increase](#)

[Contact support](#)

Browser screenshot



3. Basic Security & Networking

Infrastructure

Overview

VPC layout

Status history

Reservations

Compute

Network

VPCs

Cluster networks

Subnets

Virtual network interfaces

Access control lists

Security groups

Routing tables

Public gateways

Floating IPs

Public address ranges

Load balancers

Virtual private endpoint gateways

OverviewRulesAttached resources

Inbound rules

Search items

Refresh

Download

Settings

Create +

Name	Protocol	Source type	Source	Destination type	Destination	Value
inbound-icmp-tcp-udp-from-this-security-group	ICMP-TCP-UDP	Security group	subfloor-ladders-swampland-remix	Any	0.0.0.0/0	— :
allow-ssh	6 TCP	Any	0.0.0.0/0	Any	0.0.0.0/0	Port 22 :
allow-http	6 TCP	Any	0.0.0.0/0	Any	0.0.0.0/0	Port 80 :

Outbound rules

Search items

Refresh

Download

Settings

Create +

Name	Protocol	Source type	Source	Destination type	Destination	Value
outbound-icmp-tcp-udp	ICMP-TCP-UDP	Any	0.0.0.0/0	Any	0.0.0.0/0	— :

4. Cost Awareness

- **Check what free-tier limits exist (VM size, hours, network usage).**

IBM Cloud does not provide a free-tier VM like AWS, so my virtual server ran at approximately \$0.06 per hour. This cost stops as soon as the VM and its associated resources are deleted.

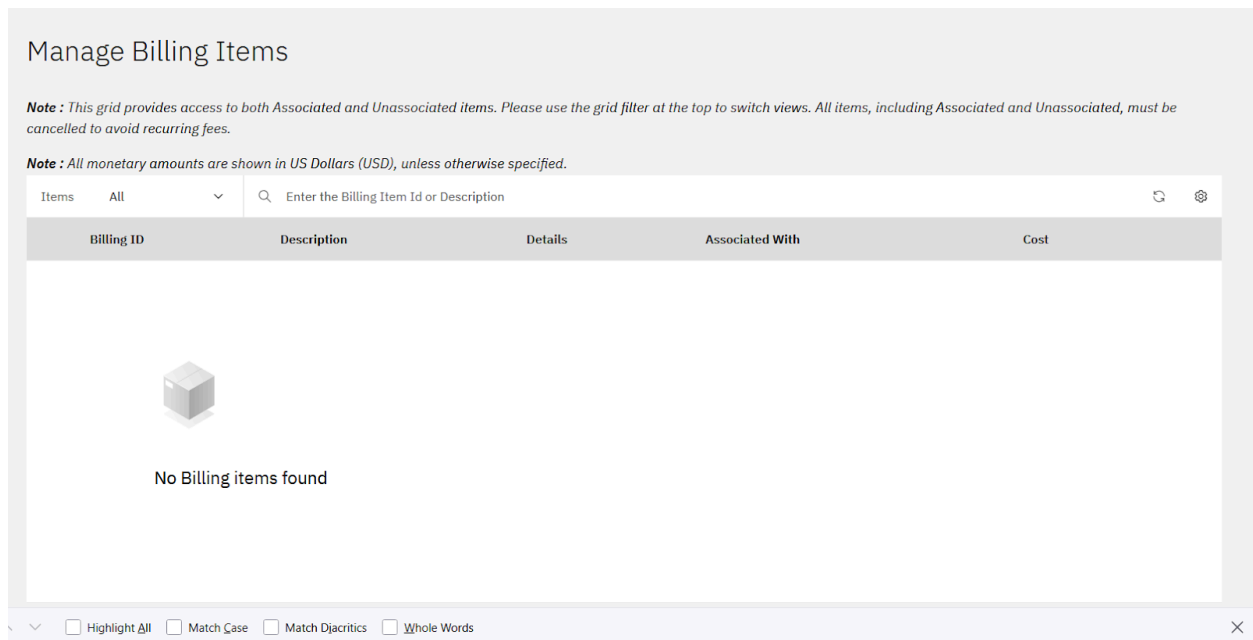
However, since my sign up I have received \$200 in credits.

- **What would be the cost if the VM kept running beyond the free tier?**

If the VM kept running beyond the free tier, IBM Cloud would begin charging based on hourly compute usage, storage, and network traffic. Costs accumulate as long as the VM and its attached resources remain active.

In this case I would pay about \$36 monthly minimum (5 cents the hour for the Flex Nano nxf-2x1 plan which was the cheapest, I used this one since there was no free option. In addition another 1 cent for the General Purpose Tier Gen2 which was Block Storage for the VPC).

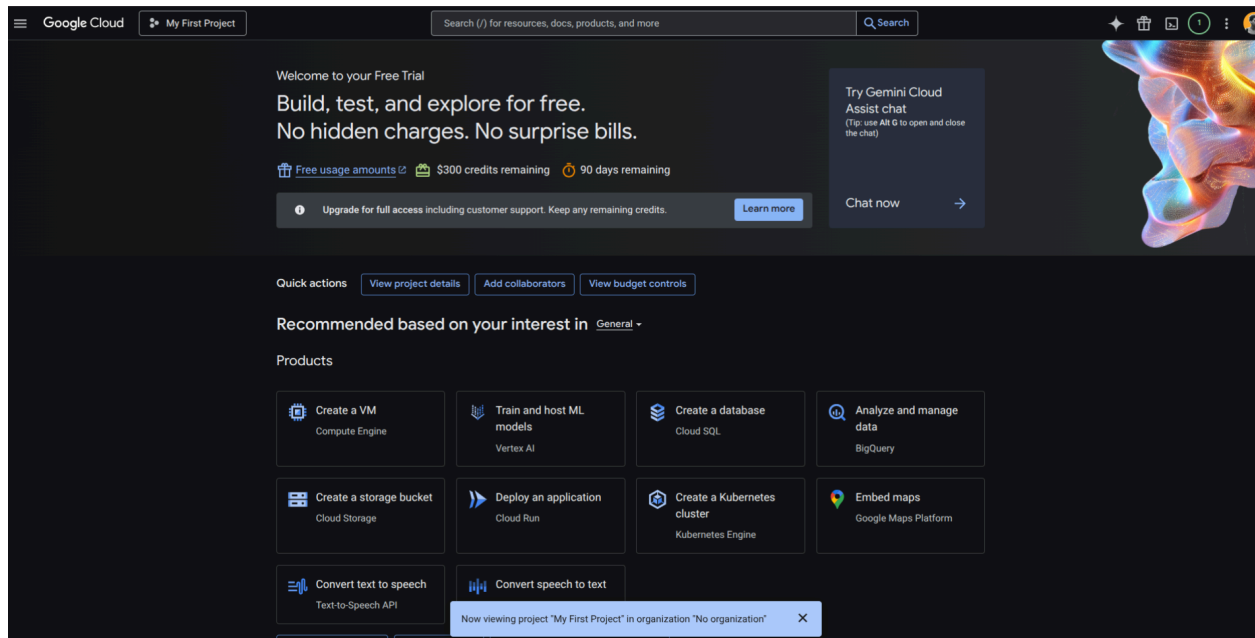
Resources Terminated:



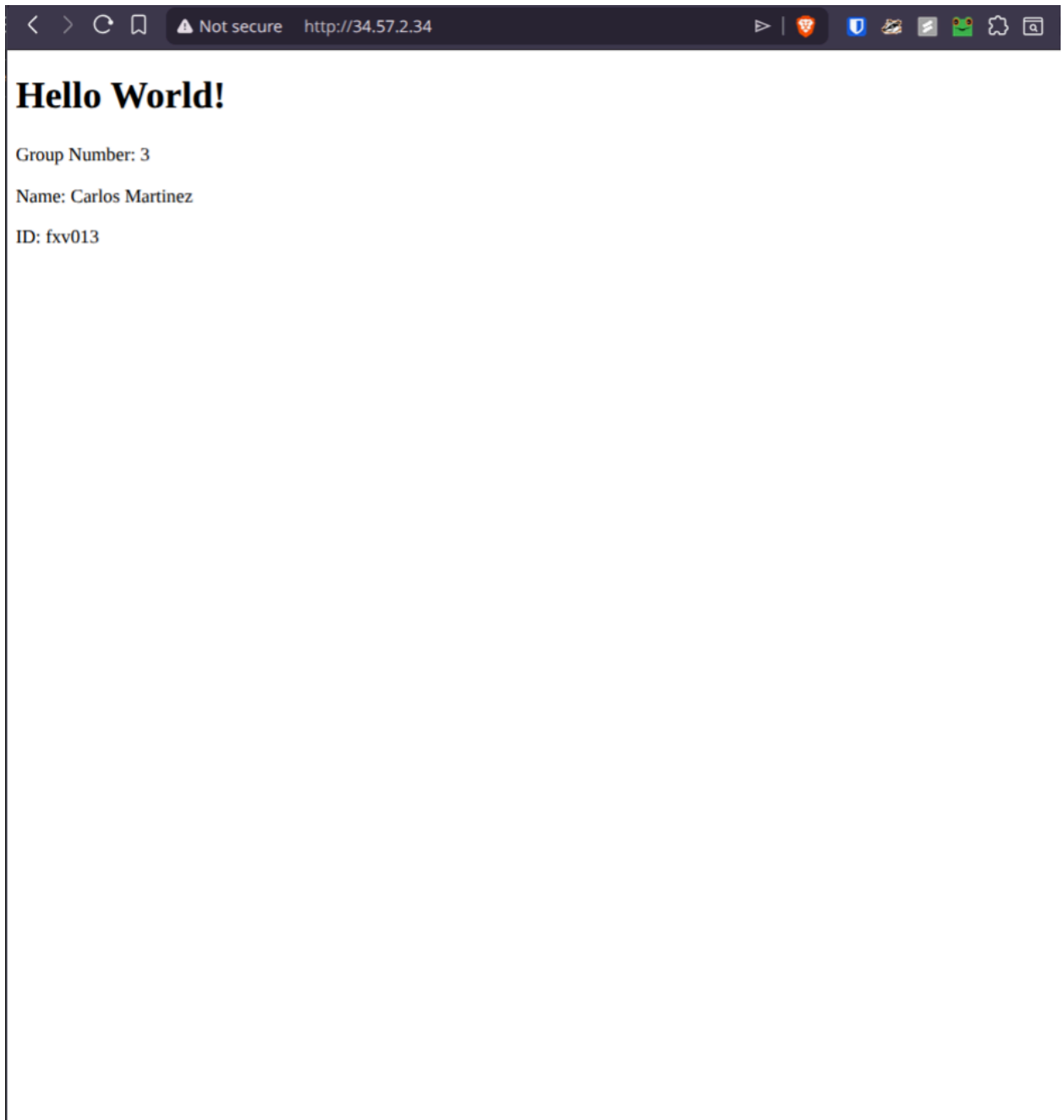
Student 3 Carlos Martinez fxv013 – [Google Cloud]

Briefly describe following based on your experience with the assigned cloud provider:

- Account setup experience: **Straight forward as far as creating the account goes. However it wouldn't let me use my abc123@my.utsa.edu email; it seems like it only accepted Gmail accounts. The UI was very simple and clean which made it easy to understand, it felt too simple and less robust which could be a pro or a con depending on knowledge**
- VM deployment steps (OS, web server, webpage): **Left Menu > Compute Engine > VM instances > Enable > Create Instance. The set up settings are right there directly where I was able to choose the name, region, zone, machine type and boot disk OS which I selected Debian/Linux 12 1Gb, the site was easier on Google Cloud than AWS.**
- Basic security setup (firewall/network rules) and its terminology: **allowed http and https access from port 80 & 443. It was pretty easy to set up. Everything was straight to the point as far as setting the firewall and enabling those ports.**
- Free tier notes and cost awareness (e.g., specific limits, what costs money beyond free tier, what monitoring/billing alert services the provider offers to avoid unexpected charges): **\$300 credit, it offers alerts if the billing exceeds a certain amount. Making this the second most generous free tier when compared with AWS.**
- Challenges encountered and resolution:
Updating the server using `sudo apt update` & `sudo apt upgrade -y` took a long time to complete. Using a different email was very complicated as it left out the ability of choosing a different one, it automatically grabbed the one that was signed in on the browser with.
- **Include Screenshots (minimum 5):**
 - 1. Cloud console after login



- 2-3. VM instance details page and Webpage running in browser



1.

- 4. Security/firewall rules

Google Cloud

My First Project

Search (/) for resources, docs, products, and more

Search

3

Network Security

Firewall policies

Create firewall policy

Create firewall rule

Learn

Cloud Armor

DDoS Dashboard

Cloud Armor policies

Adaptive Protection

Cloud Armor Service Tier

Cloud IDS

IDS Dashboard

IDS Endpoints

IDS Threats

Cloud NGFW

Dashboard

Firewall policies

Threats

Firewall endpoints

Secure Access Connect

Realms

Preview

Attachments

Preview

Cloud NSI

Deployment groups

Endpoint groups

Secure Web Proxy

SMTP port 25 disallowed in this project. [Learn more](#)

Refresh

Configure logs

Delete

Filter

Enter property name or value

	Name	Type	Targets	Filters	Protocols / ports	Action	
☐	default-allow-http	Ingress	http-server	IP ranges:	tcp:80	Allow	▼
☐	default-allow-https	Ingress	https-server	IP ranges:	tcp:443	Allow	▼
☐	default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	▼
☐	default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	▼
☐	default-allow-rdp	Ingress	Apply to all	IP ranges:	tcp:3389	Allow	▼
☐	default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	▼

Network firewall policies

Firewall policies let you group several firewall rules so that you can update them all at once, effectively controlled by Identity and Access Management (IAM) roles. [Learn more](#)

Refresh

Filter

Enter property name or value

1.

- 5. Confirmation of resource termination

Google Cloud

My First Project

Search (/) for resources, docs, products, and more

Search

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Compute Engine

VM instances

Create Instance

Import VM

Refresh

Learn

Overview

Security risk overview

Virtual machines

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Committed use discounts

Reservations

Migrate to Virtual Machines

Storage

Disks

Storage Pools

Snapshots

Images

Async Replication

Instance groups

Instance groups

Marketplace

Release Notes

Instances

Observability

Instance schedules

VM instances

Filter Enter property name or value

Status

Name ↑

Zone

Recommendations

In use by

Internal IP

Connect

VM Instances

Compute Engine lets you use virtual machines that run on Google's infrastructure. Create micro-VMs or larger instances running Debian, Windows, or other standard images. Create your first VM instance, import it using a migration service, or try the quickstart to build a sample app.

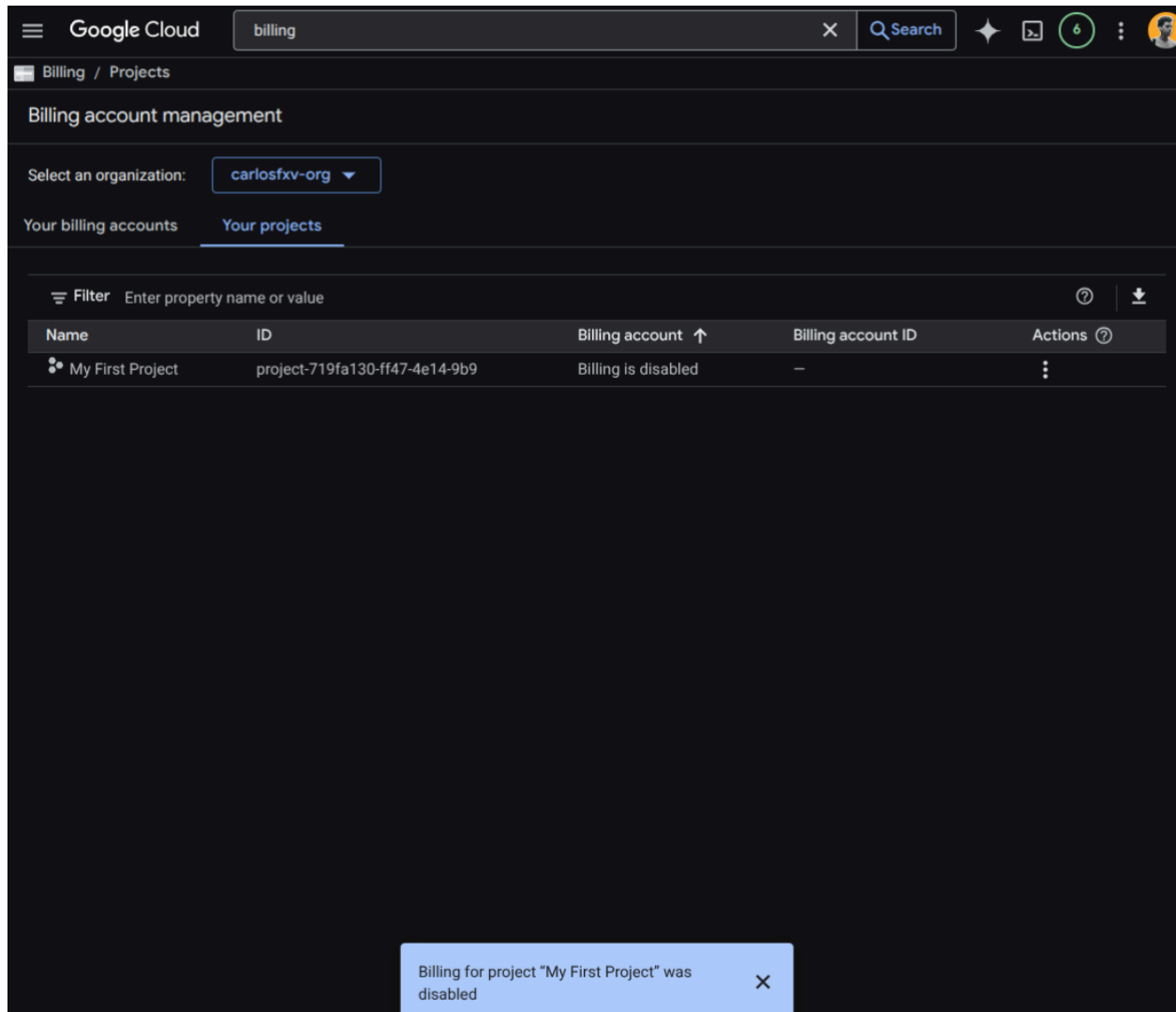
Create instance

Take the quickstart

Instance deleted

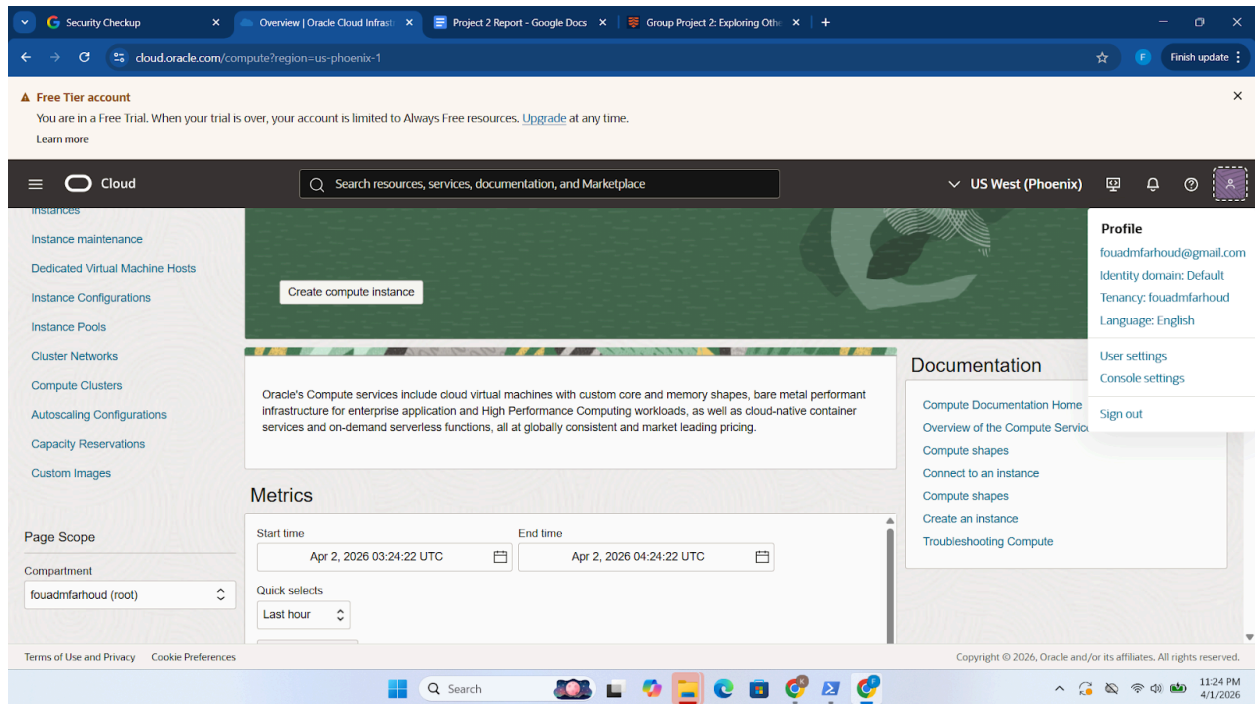
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1.



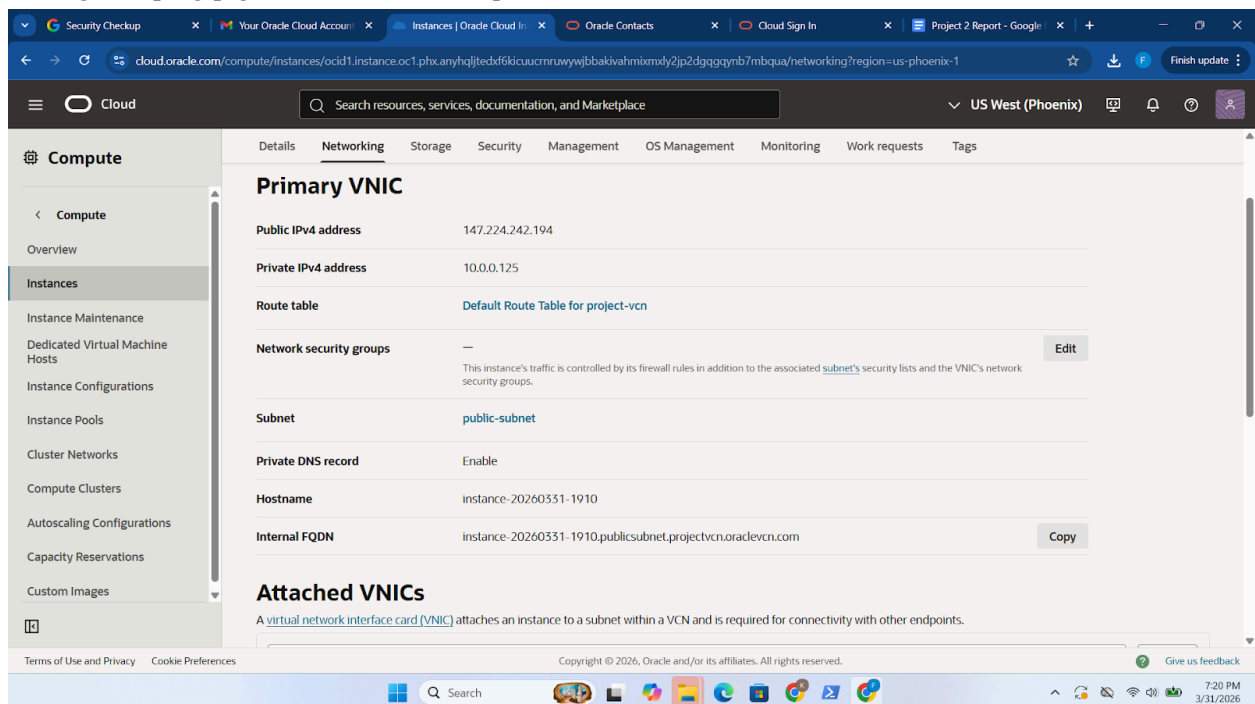
Student 4 Fouad Farhoud and idu077 – [Oracle Cloud]

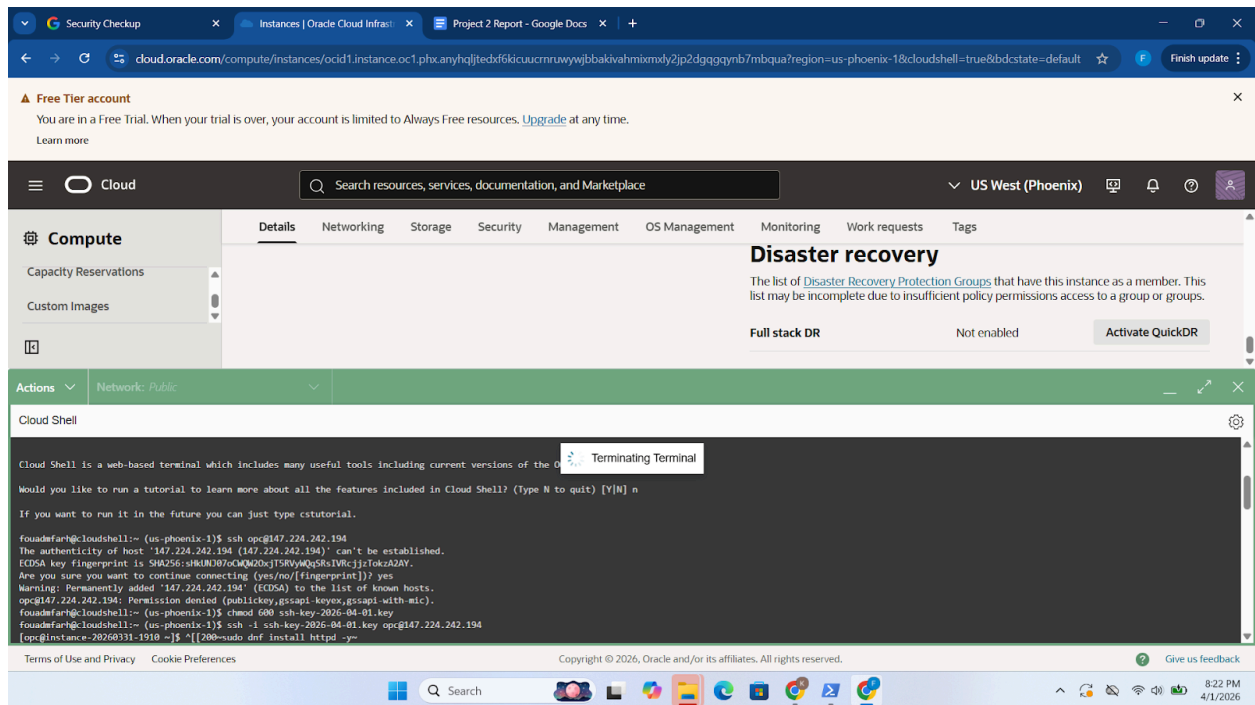
Account setup - Overall I had a pretty smooth experience when it came to setting up my account with Oracle cloud. I will say that the website was pretty slow and lagged a lot, which leads me to my other issue whenever it told me my account was created I tried to log in but it was not working due to the email confirmation confirming my account was created was not sent out yet which came 15 minutes later so I was super confused on that. I was charged \$1 for a credit card confirmation



1. VM Deployment and a static web page

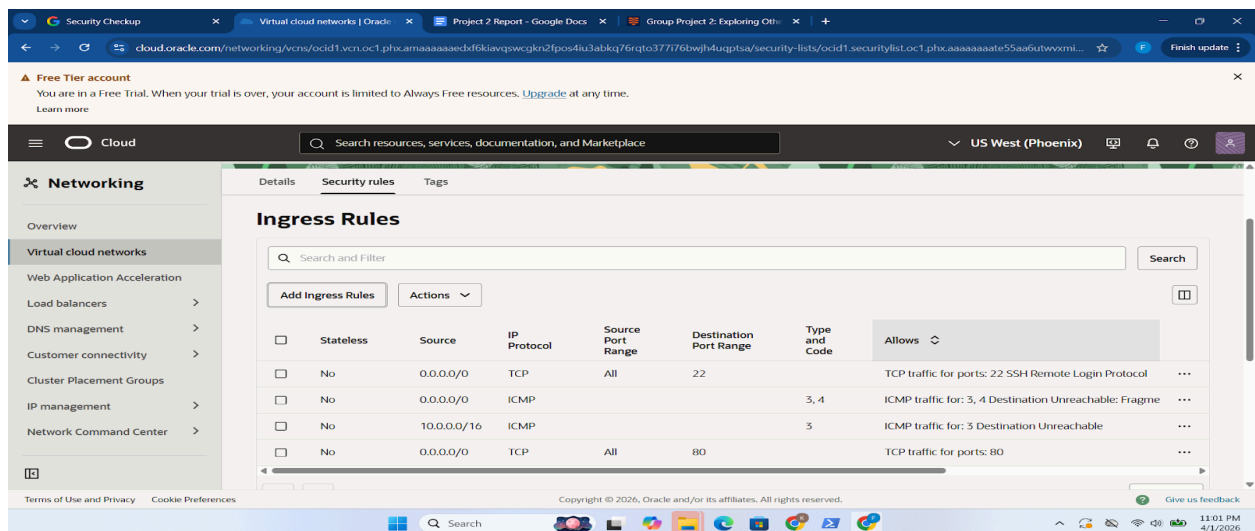
This screenshot shows the successfully deployed virtual machine instance along with the assigned public IP address/URL. The presence of the public endpoint confirms that the VM is running and accessible over the internet. I got the download to happen for me to run a Apache server but the terminal kept saying waiting for “ping/pig” to install with a sequence of numbers





Basic Security & Networking -

This screenshot demonstrates that the virtual machine's security rules were configured to allow inbound HTTP (port 80) and HTTPS (port 443) traffic. These firewall settings are essential for hosting a public-facing website, as they permit browser-based requests from external users. At the same time, limiting access to only these ports helps reduce unnecessary exposure and follows basic security best practices.



Cost Awareness - Oracle Cloud offers an Always Free tier that includes limited resources such as small virtual machine instances, a set number of compute hours per month, and restricted network usage. As long as usage stays within these limits, no charges are incurred. If the VM runs beyond the free tier, it may cost approximately \$0.01–\$0.02 per hour. This results in about \$0.24–\$0.48 per day, or roughly \$7–\$15 per month if left running continuously. Oracle Cloud provides tools such as Budgets and Cost Analysis to help users monitor spending and avoid unexpected charges. The Budgets feature allows users to set a spending limit and receive alerts (via email) when usage approaches or exceeds that limit. Oracle charges based on services you are using I terminated all services even though those services are already free from what I was using it for.

The screenshot shows the Oracle Cloud account page. At the top, there's a navigation bar with the Oracle Cloud logo and a search bar. Below the navigation bar, there's a section for 'Universal Credits' with a large green 'S' icon and the word 'ACTIVE' below it. To the right of the icon, there's a box containing subscription information:

Subscription information		
Subscription ID: 76880538	Original start date: Mar 31, 2026 UTC	Total commitment value: \$0.00
Type: Infrastructure	Start date: Mar 31, 2026 UTC	Currency: USD
CSI number: 124284734	End date:	

Below the subscription information, there's a 'Billing schedule' section. It has a 'Download as CSV' button and a table with the following columns: Status, Invoice date (UTC), Start date (UTC), End date (UTC), Frequency, Quantity, Net unit price, and Line net amount. The table is currently empty, and a message below it states: 'There are no billing schedules available for the selected service line'.

At the bottom of the page, there's a footer with 'Terms of Use and Privacy' and 'Cookie Preferences' links. The bottom of the browser window shows the Windows taskbar with various icons and the system clock indicating 2:48 PM on 4/2/2026.

3. Group Reflections

- Which provider was easiest for beginners to navigate and use? Why?

Google Cloud was the easiest for beginners to navigate and use. The terminology, layout, and UI felt really intuitive, and everything needed to create an instance was on one page, which made setup simple. At the same time, I could see how more advanced users might find it a bit limiting because of how simplified it is

- Which provider offered the most generous free tier, and which felt most limited or had unexpected complexities?

Google Cloud had the most generous free tier, offering \$300 in credits for 90 days. Compared to that, Microsoft Azure felt more restrictive, with a less flexible free tier and more unexpected complexity when setting things up.

- **What were the most significant differences and similarities you observed between these providers and your experience with AWS in terms of workflow, terminology, and conceptual models?**

In terms of workflow and conceptual models, Microsoft Azure and Google Cloud felt the most similar, while IBM Cloud felt closer to AWS in structure. Across all providers, the core concepts such as compute, storage, and networking remain consistent, and all platforms offer web consoles, CLI tools, and APIs. However, the user interfaces and service names vary significantly, which can make switching between providers confusing.

- **What terminology differences did you find most confusing (e.g., Security Groups vs. Firewall Rules vs. NSGs)? How might this affect one switching between providers?**

The most confusing differences were in networking and security terminology. For example, AWS uses Security Groups, Google Cloud uses Firewall Rules, Microsoft Azure uses Network Security Groups (NSGs), and IBM Cloud uses Security Groups and ACLs. While these all serve similar purposes, they are applied differently. In Google Cloud, Firewall Rules use tags for indirect targeting rather than attaching directly to instances, while in Azure, NSGs can be applied at multiple levels such as subnets or network interfaces. This can lead to confusion and misconfiguration when switching providers.

- **Based on your experience, which provider would your group consider using again for a personal project or a small startup, and what are the reasons for your choice?**

Based on our experience, we would choose Google Cloud for a personal project or small startup. It provides a user-friendly interface, a straightforward workflow, and a generous free tier, making it ideal for beginners and smaller-scale projects. However, for larger-scale applications, we would consider transitioning to Microsoft Azure due to its more robust enterprise features and scalability.

- **Describe the most significant technical or collaborative challenge your group encountered during this project, and how you collectively addressed it.**

The most significant challenge our group faced was coordinating meeting times due to midterms and other exams. For example, we initially planned to meet on a Saturday, but scheduling conflicts caused us to miss it. To address this, we rescheduled and held a meeting the following Monday to assign providers, and then met again on Wednesday to complete the project. This improved communication helped us stay on track and finish the assignment.

4. References

List the key resources used (official docs, tutorials, GenAI tools, etc.).

<https://docs.cloud.google.com/docs>

IBM Cloud:

<https://cloud.ibm.com/docs/vpc>

Microsoft Copilot. (2026). For Interactive guidance and explanations for IBM Cloud VPC

Azure Cloud:

[Azure documentation | Microsoft Learn](#)